

Session 6: Case Study on AI-Powered Disaster Damage Assessment

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*This activity
is supported by:*

The NATO Science for Peace
and Security Programme



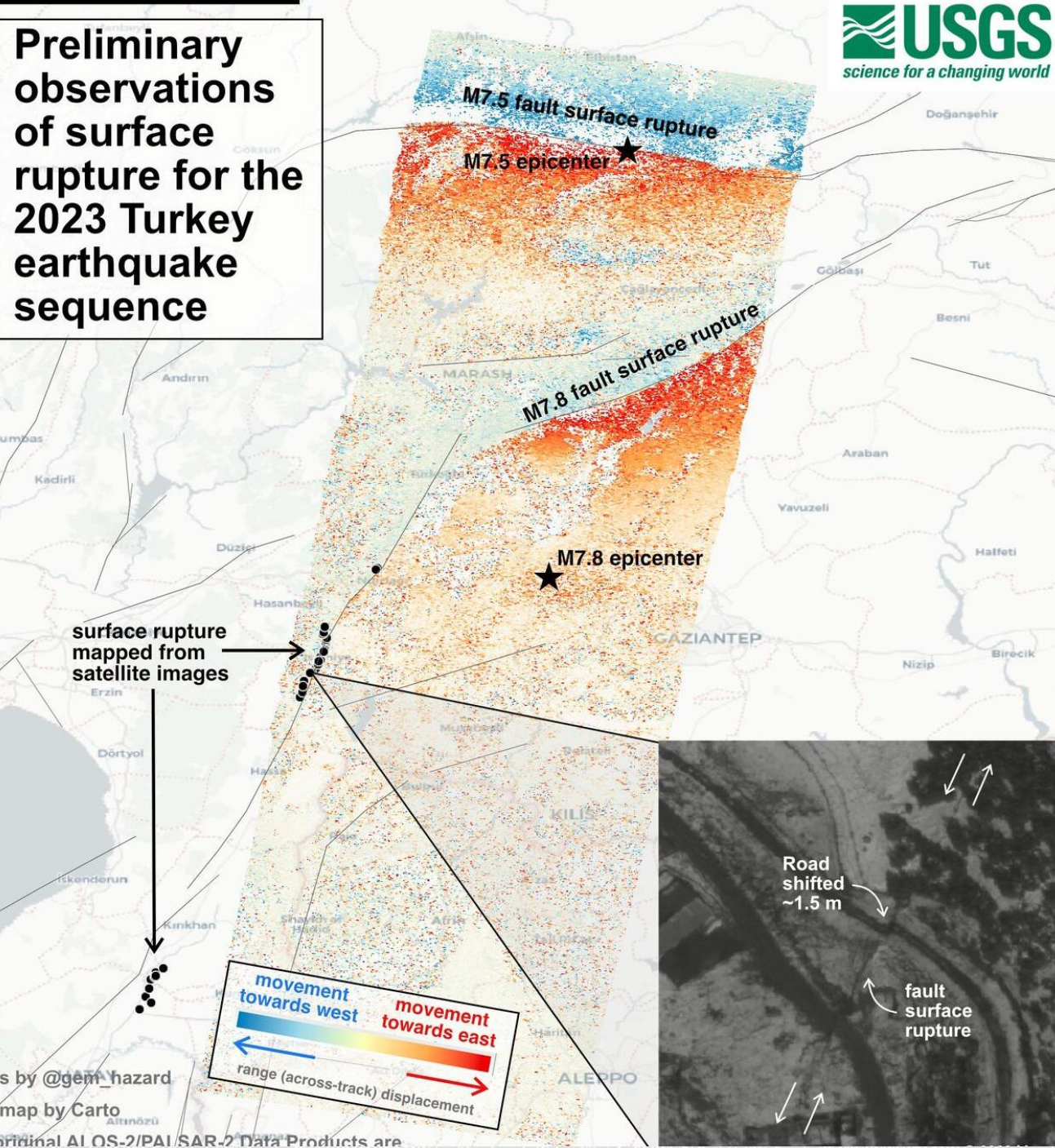


Agenda

- UCF** Opening - The crisis and urgency
- UCF** Why AI? How can AI help?
- UCF** AI Model Basic Components
- UCF** Data - Maps
- UCF** The CNN Solution - How it works
- UCF** Impact - Speed, resources, prioritization, long-term use, challenges
- UCF** Looking Forward - Progress, global capacity
- UCF** Basic Implementation
- UCF** Wrap-up
- UCF** Q&A

**Figures adapted from different resources for illustration only.
References will be included in the last slide.**

Preliminary observations of surface rupture for the 2023 Turkey earthquake sequence



2023 Turkey–Syria Earthquakes: Sequence at a glance

- Local time 04:17 (TRT), 6 Feb 2023 — A very large M7.8 earthquake hit southern Turkey.
- Within ~9 hours: a second large event M 7.5 to the north; many strong aftershocks.
- Shaking intensity reached XII (Extreme) in parts of Antakya and near the epicentral area.
- Affected regions span 11 provinces in Turkey and north-western Syria.
- 50,000+ deaths across Turkey and Syria
- 84,000+ buildings damaged or destroyed
- ~14 million people affected
- ~ \$34.2 billion USD direct economic losses

The Challenge?

In the critical first 72 hours, how do you assess damage across 30+ cities?

How can you prioritize rescue operations without accurate, real-time damage data?

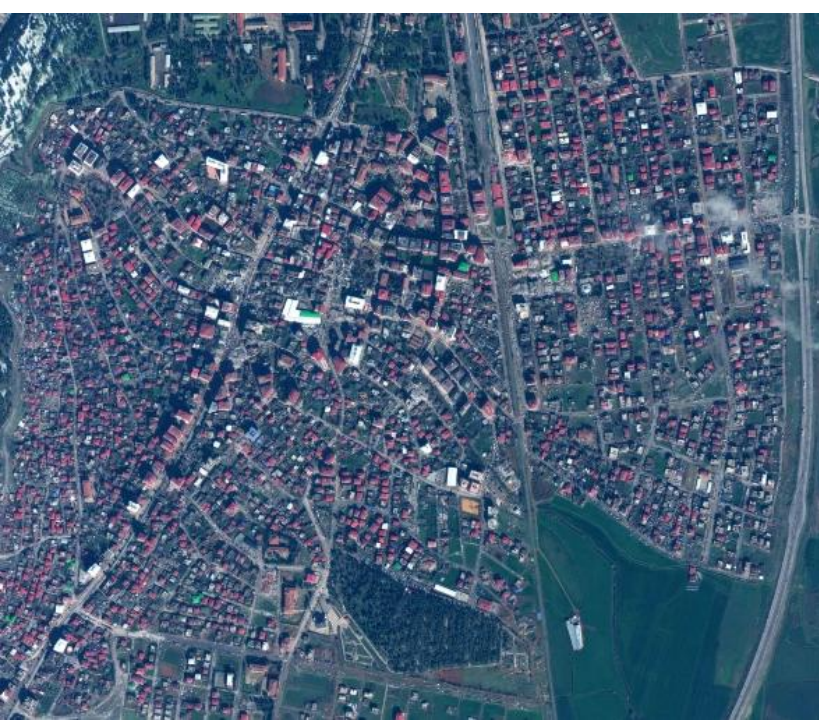
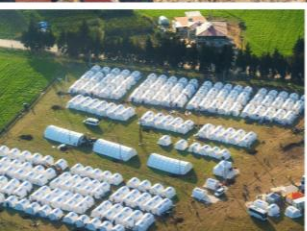
How do you assess structural integrity of thousands of buildings without physical access?

How can you estimate the number of displaced people when roads and transportation are blocked?

How do you coordinate multiple agencies when information is fragmented and outdated?

How can you ensure effective resource allocation across dozens of affected regions?



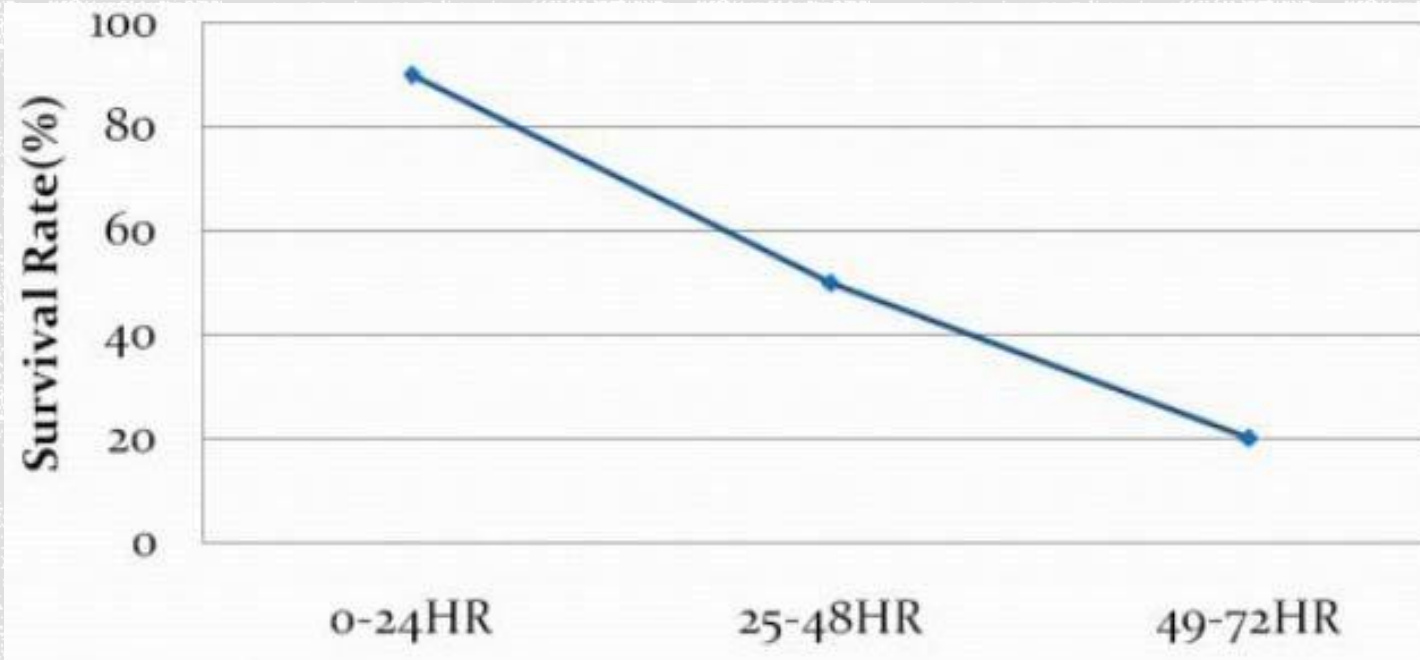


WHY AI?

- Is AI the game-changer we need?
- How can AI help save lives?
- What role can AI play in disaster response and recovery?

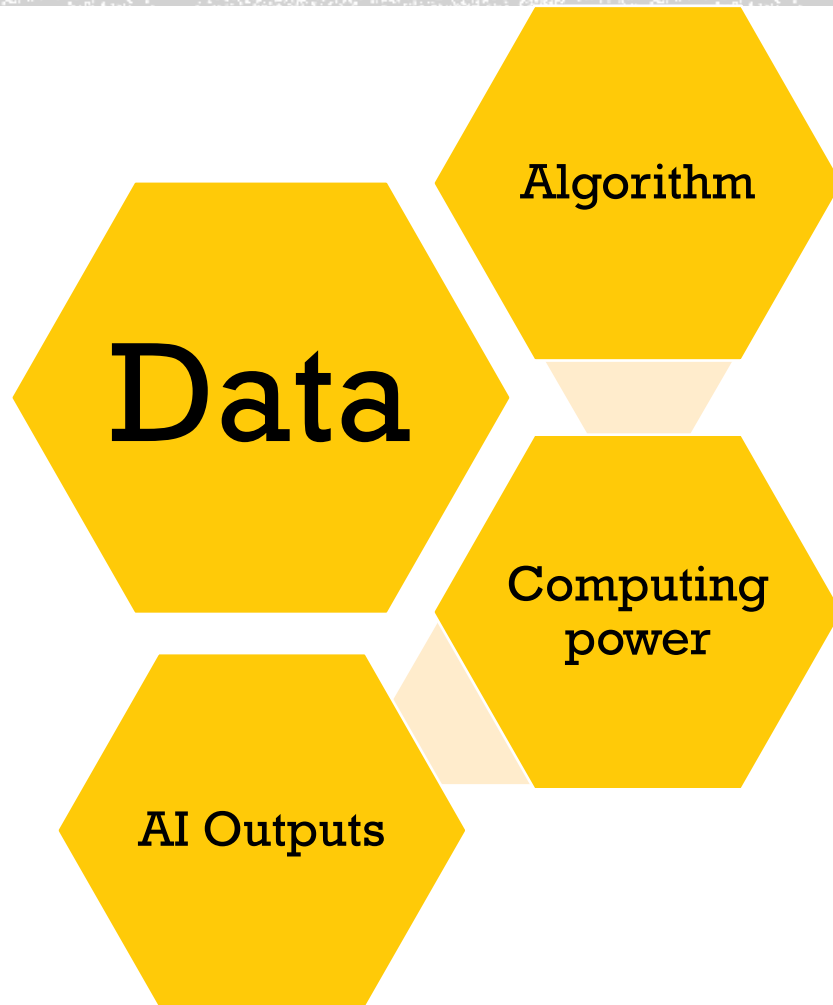
IT SAVES TIME

WHY SPEED MATTERS: THE 72-HOUR WINDOW

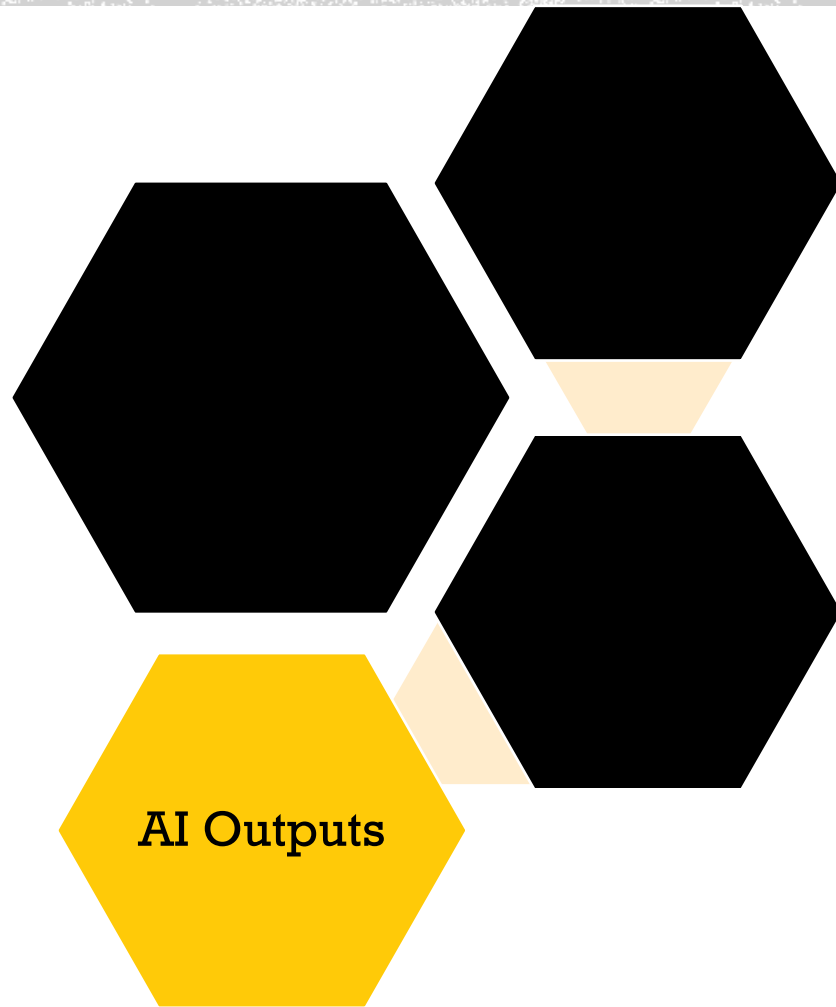


- ❖ First 24 hours: 90% survival rate
- ❖ 24-48 hours: 90% - 50% survival rate
- ❖ 48-72 hours: 50% - 20%
- ❖ After 5 days: Below 10%

Receiving damage maps by Days 1-3 instead of Day 10 allows rescue teams to prioritize efforts while the majority of trapped individuals are still alive



**COMPONENTS
OF ANY AI
MODEL?**



COMPONENTS
OF ANY AI
MODEL?

What are the **most critical outputs from an AI model** during disaster response? **Damage Maps**



NATO Response: why rapid maps matter on the ground

UCF Allies set up temporary housing in Hatay and supported an air-bridge for relief supplies.

UCF NATO's support helped house displaced families and move tents, blankets, and medical supplies quickly.

UCF Rapid damage maps help choose where shelters go and which routes are safe for deliveries.

UCF Damage maps guide Search and Rescue teams to the areas that need the most attention.



COMPONENTS
OF ANY AI
MODEL?

What **type of data** does AI need to assess disaster damage (to get damage maps)?



Satellite Imagery Sources for Disaster Assessment

Main Types of Satellite Imagery:

- **Optical** (Visual Images): Shows roads, buildings, vegetation.
 - First used: 1972 (Landsat 1).
- **SAR** (Synthetic Aperture Radar): Uses radar waves, works in all weather and at night.
 - First used: 1990s for floods and earthquakes.
- **Multispectral**: Captures multiple light bands (visible + infrared) for land and water analysis.
 - First used: 1972 (Landsat missions).



Satellite Sources Used in Turkey Earthquake

Key Satellite Sources and Their Roles:

- **Sentinel-1 (SAR):** to detected ground changes and create damage proxy maps.
- **ALOS-2 (SAR):** to estimated building damage ratios using radar polarization.
- **Planet Labs & Maxar (Optical):** High-resolution images for building-level damage classification.
- **WorldView (Optical):** Sub-meter resolution for mapping surface ruptures and fault lines

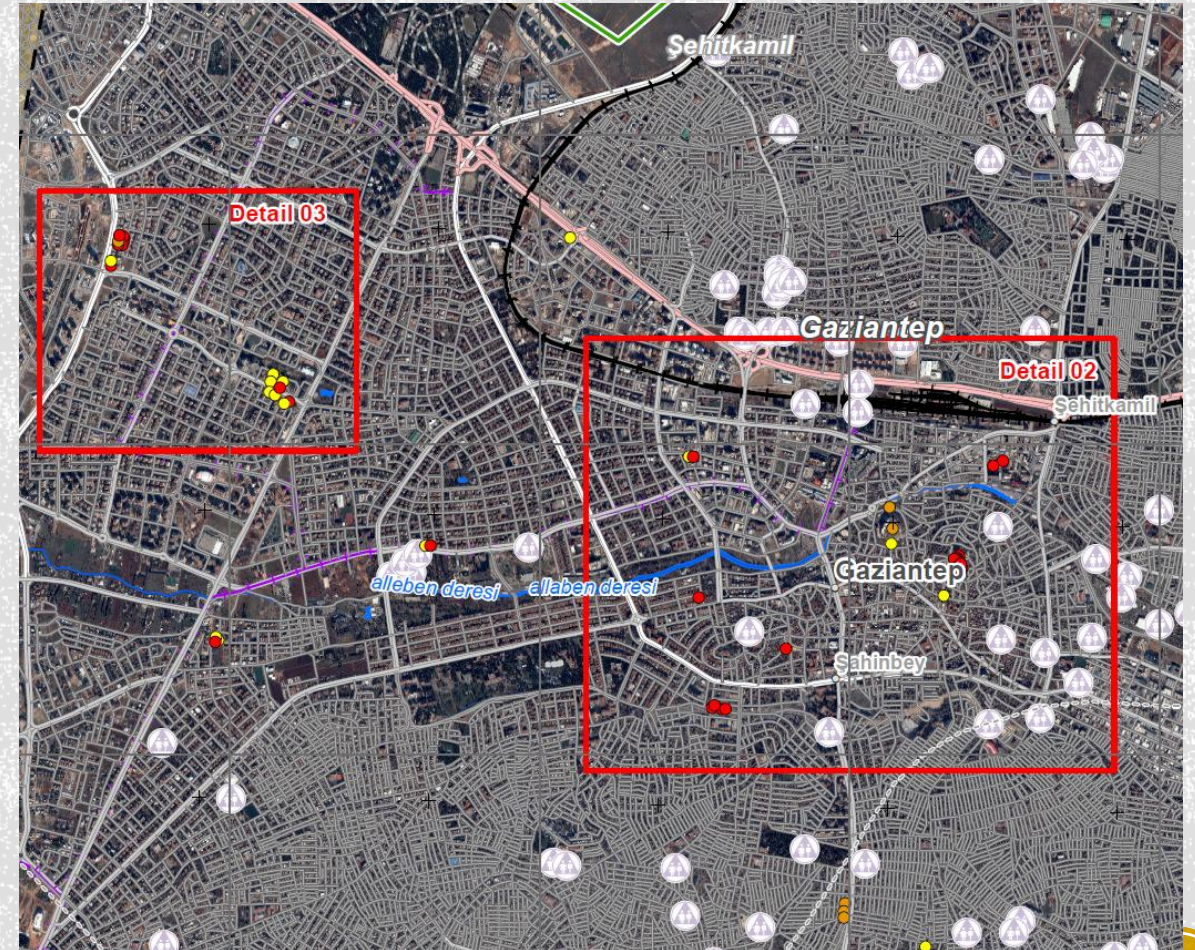
AOIs & Impact Scale

Emergency mappers in Europe (Copernicus EMS) and the UN Satellite Centre (UNOSAT) were activated quickly to make damage maps from satellite images.

They created multiple mapped areas (“AOIs”).

Examples of mapped places: Kahramanmaraş, Antakya, Nurdağı, Elbistan.

AOI (Area of Interest): A selected zone on the map where teams focus analysis.



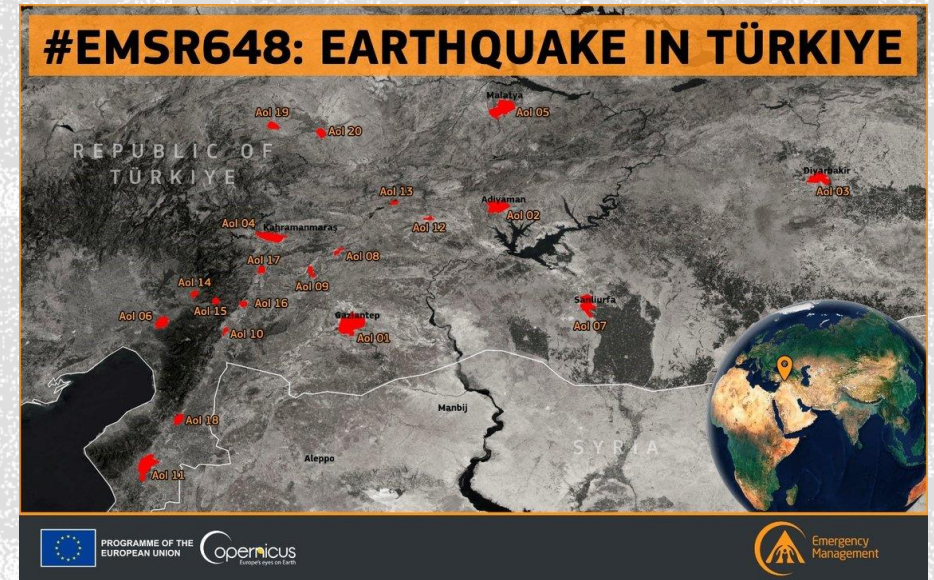
Authoritative Maps Responders Actually Use

Copernicus EMS “Grading” (EU):

- Classifies damage level fast (typically at urban-block scale)
- Designed for rapid decisions in the first days.

UNOSAT “Comprehensive Damage Assessment” (UN):

- Building-by-building status over selected cities;
- Higher detail
- Iterative updates as imagery improves.



Grading map: a product that classifies extent of damage to help set priorities.

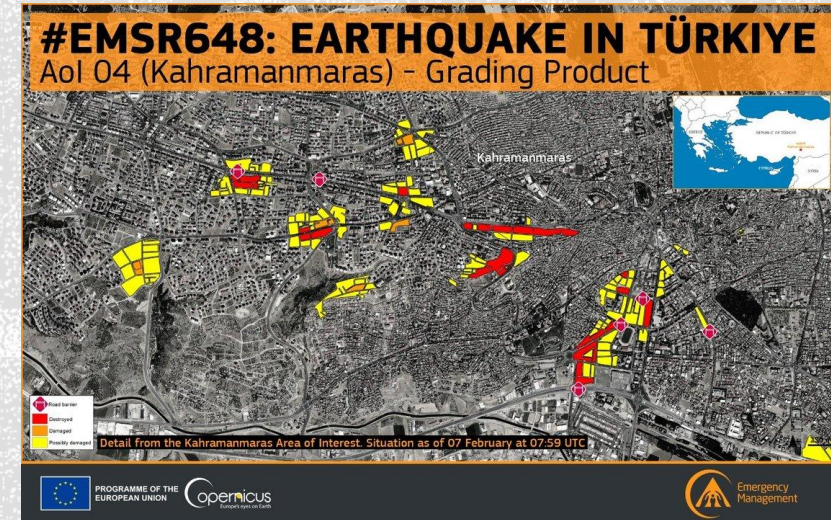
Why speed vs accuracy matters

Early “Grading” (fast)

<u>Pros</u>	<u>Watch-outs</u>
Hours–days	Not every damaged building detected
Good enough to prioritize cities/quarters	Class thresholds can be conservative.

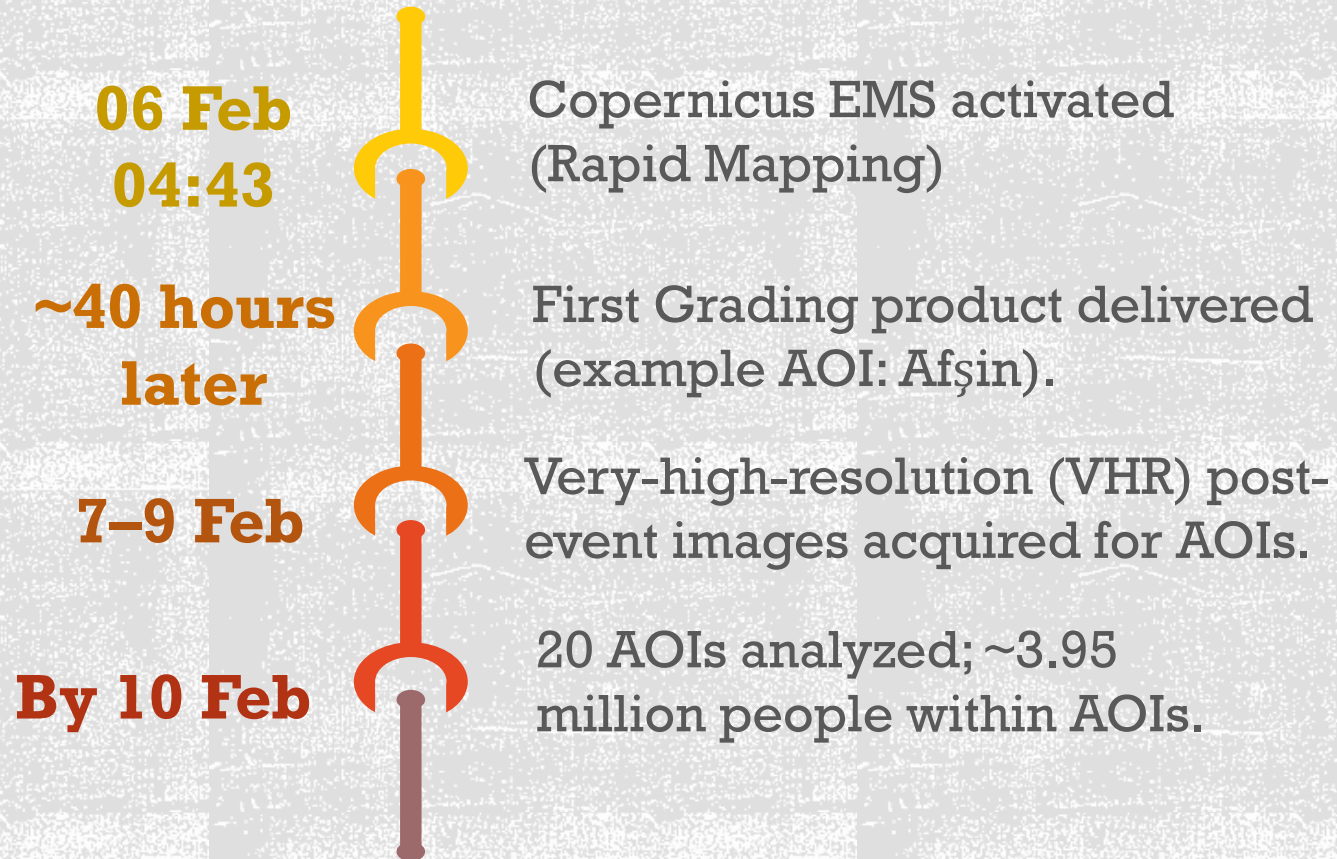
Comprehensive DA (richer)

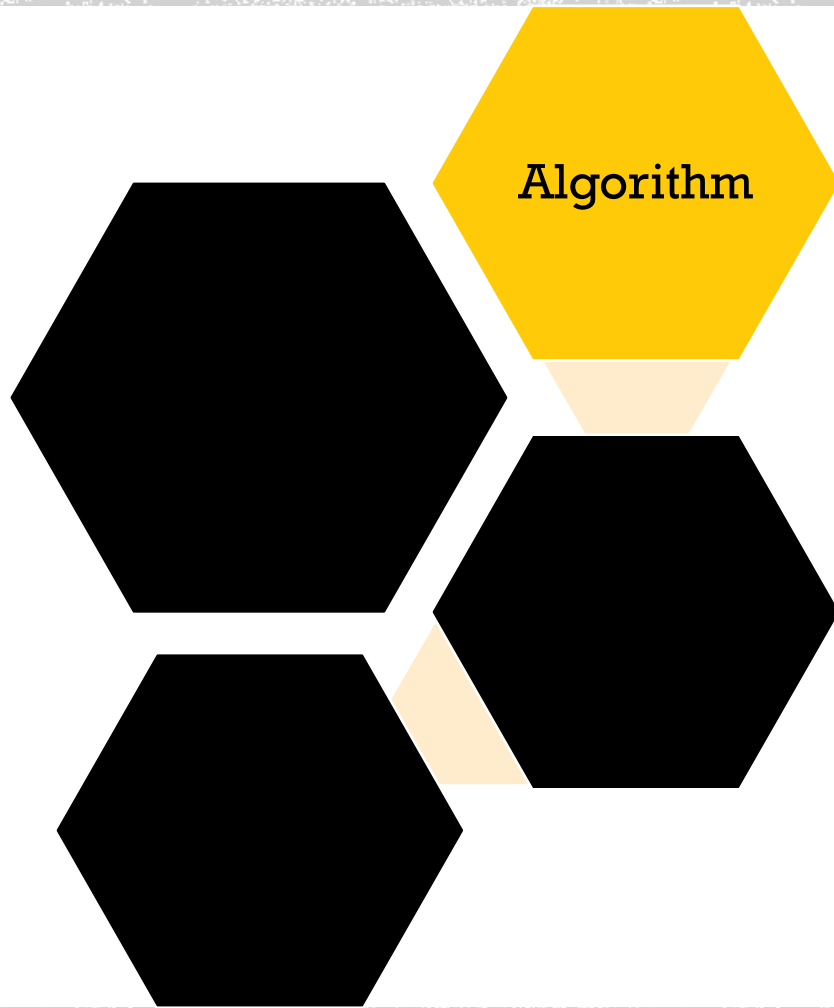
<u>Pros</u>	<u>Watch-outs</u>
Finer building counts	Arrives later
Better for resource planning and damage statistics.	May miss very recent changes if imagery lags



**Early maps get us moving;
later maps get us precise**

How fast the official maps arrived





COMPONENTS
OF ANY AI
MODEL?

What **algorithms** power AI to generate accurate damage maps?

What AI models typically ingest



Source image

Inference visualization



- Typical pipelines pair **“before” and “after” images** of the same AOI.
- This enables **change-focused analysis** (e.g., collapse, rubble, roof loss).
- Inputs are usually **very-high-resolution imagery**, similar to what Copernicus EMS and UNOSAT use in their products.

How CNNs Assessed Building Damage: A Two-Step Process

STEP 1: Find the Building

- Satellite Image
- CNN Localization
- Building Outlines

STEP 2: Assess the Damage

- Each Building Image
- CNN Classification
- Damage Level (0-3)

Everything you learned this week (Sessions 1-5) was applied here in a real emergency

Multiple Models Working Together

1. Building Detection

- U-Net (segmentation network)
- Specialized for finding object boundaries
- To draw outlines around buildings

2. Damage Assessment

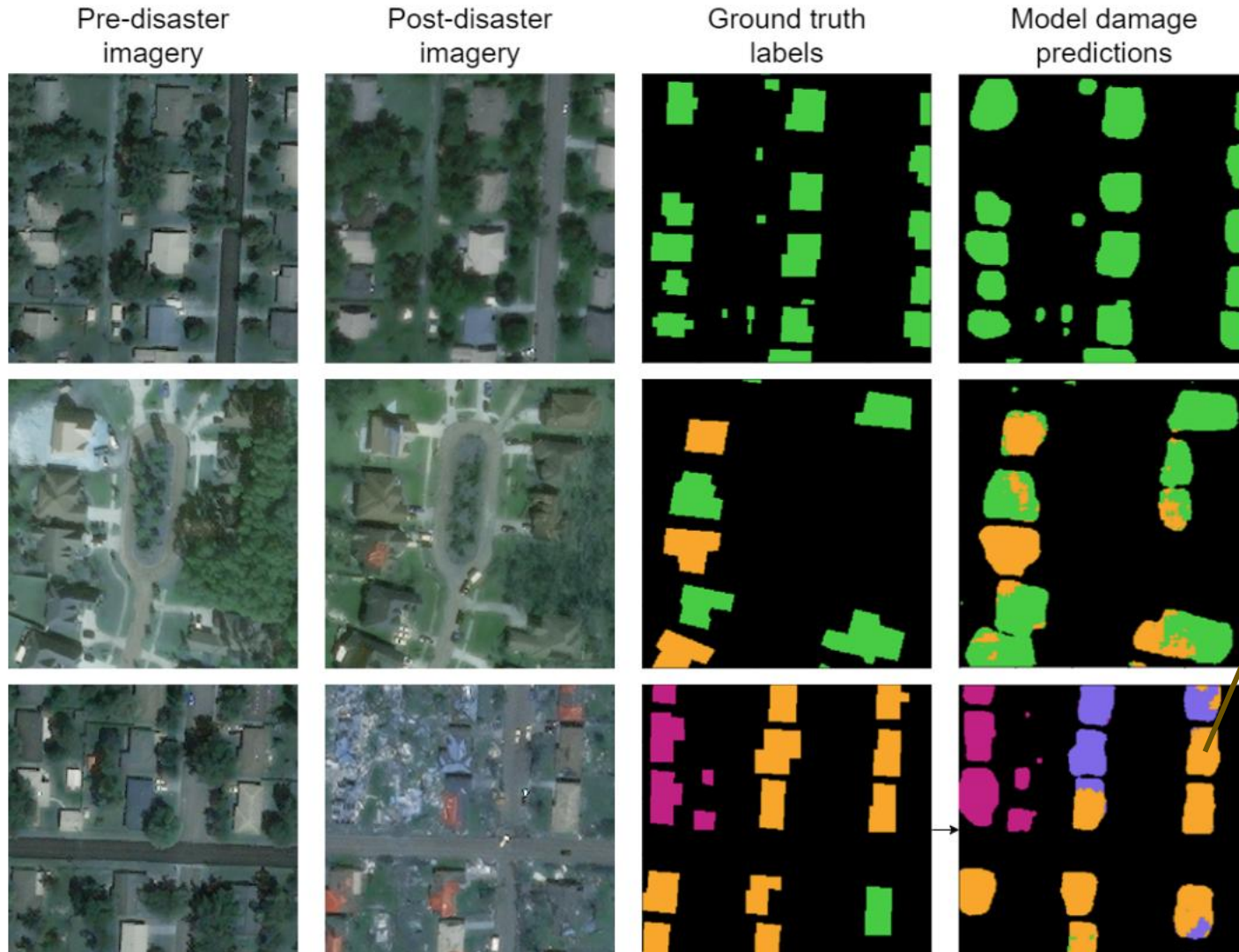
- ResNet + Others
- Pre-trained on thousands of past disasters
- To classify damage for each building

3. Ensemble Strategy

- Used 3 different models voting together
- If they disagree → flag for human review
- If they agree → trust the prediction

**Why multiple models? Single model = ~75% accurate.
Three models voting together = ~85% accurate.**

The pipeline: From buildings to damage grades



- Big images (AOIs) are **patched** into small tiles for fast processing.
- **Building footprints** define the units we care about.
- A **change-aware segmenter** analyzes pre vs post pixels.
- Outputs are **aggregated** per building into a 4-level damage grade

- **No damage** — building intact
- **Minor** — limited visible damage (e.g., small roof loss)
- **Major** — heavy damage (partial collapse)
- **Destroyed** — near-total or total collapse

Why change detection (vs single-temporal)

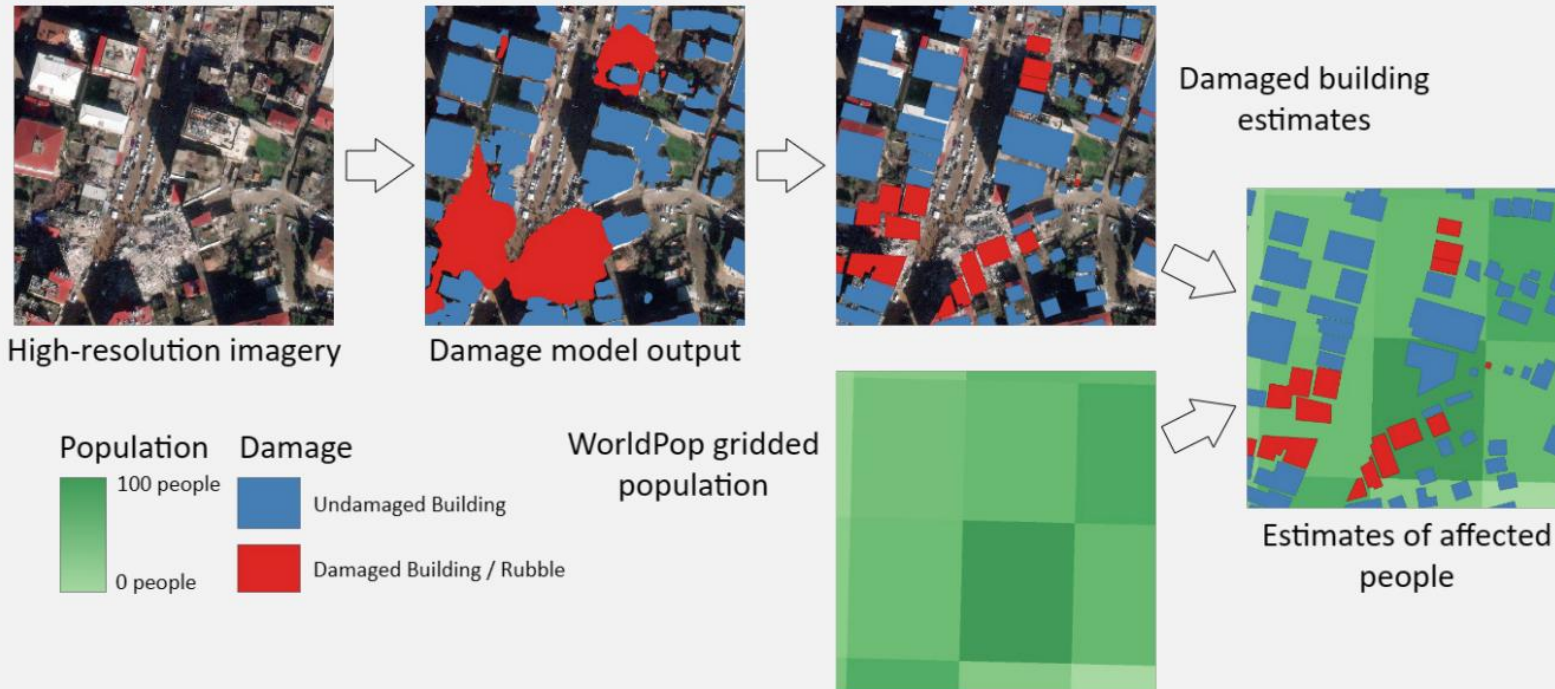


[Online Maxar](#)

- **Single-temporal** (post-only) can confuse damage with pre-existing conditions.
- **Change-detection** (pre+post) reduces false alarms by focusing on what changed.
 - Works well when look angle and illumination are reasonably matched.

Change-aware inputs tend to be more informative

From damage maps to estimates of affected people



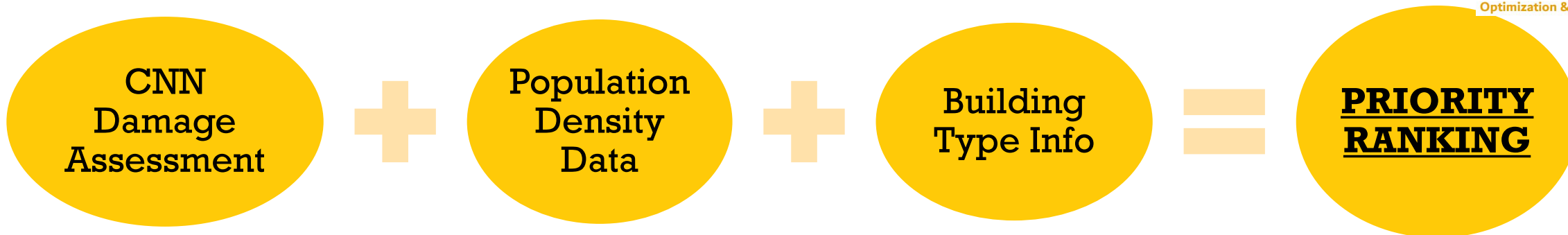
- **Input imagery**: pre/post very-high-resolution satellite images over the same AOI.
- **Model output** (published methods): per-building damage classes (e.g., undamaged vs damaged/rubble).
- **Population layer**: WorldPop gridded counts (e.g., people per cell).
- **Overlay**: intersect damaged buildings with gridded population to estimate affected people near damage clusters.

Assumptions/limits:

- **time-of-day effects**
- **residential vs non-residential use**
- **population grid resolution & recency**

Uncertainty should be communicated.

Prioritization - Not All Damaged Buildings Are Equal



Building Damage	Population Density	Priority Level
Destroyed	High (apartments, hotels)	● HIGHEST
Major Damage	High	● HIGHEST
Destroyed	Low (warehouses, empty)	● Medium
Major Damage	Medium	● Medium
Minor Damage	High	● Lower
No Damage	Any	● No Action

Search and rescue teams had a ranked list within 48 hours.
 Focused on apartment buildings and hotels first—highest probability of trapped persons.

From Earthquake to Damage Maps: The 72-Hour Timeline

Feb 6, 4:17 AM (Day 1)

🌟 EARTHQUAKE STRIKES

↓ 36 hours

Feb 7, 4:00 PM (Day 2)

🛰️ First satellite images received

↓ 4-6 hours

Feb 8 (Day 3)

💻 Preprocessing (cleaning images, aligning coordinates)

↓ 2-4 hours

Feb 8-9 (Days 3-4)

🤖 CNN Processing

- Step 1: Find all buildings (3-4 hours)
- Step 2: Classify damage (2-3 hours)

↓ 4 hours

Feb 9 (Day 4)

👥 Human review of uncertain cases (~12% of buildings)

↓ 2 hours

Feb 9 (Day 4)

🗺️ Create maps, distribute to rescue teams

TOTAL: ~72-96 hours from earthquake to maps

Traditional VS CNN-Assisted

Method	Time to City-Wide Assessment
Traditional (manual)	7-14 days
CNN-Assisted	3-4 days
Time Saved	4-10 days

80% of earthquake survivors are rescued in the first 72 hours. Getting damage maps on Day 3 instead of Day 10 saves lives



VALIDATION RESULTS: WHERE THE CNN GOT IT RIGHT AND WRONG

- **CNNs were best at extremes** (intact or destroyed) and struggled with subtle damage (minor).
- **High Confidence** (>80%) → Trust CNN, **deploy resources**
- **Low Confidence** (<60%) → **Human expert reviews** before deployment.
- **12% of buildings flagged for human review** caught most potential errors

CLASS 0	CLASS 1	CLASS 2	CLASS 3
No Damage	Minor	Major	Destroyed
89%	53%	68%	82%
✓	⚠	⚠	✓
Excellent	Struggled	Moderate	Excellent

RESOURCE ALLOCATION - DIRECTING HELP WHERE IT'S NEEDED

Without CNN Assessment (Day 1-2)

Resource Distribution:

-  Rescue teams spread uniformly
-  Medical units distributed evenly
-  Helicopters searching systematically

Problems:

-  Many teams in low-damage areas
-  Insufficient resources in high-damage zones
-  No prioritization system

“We were searching everywhere but didn't know where to focus.”— Turkish Search and Rescue Team

With CNN Assessment (Day 3+)

Resource Distribution:

-  75% of teams concentrated in 47 high-damage zones
-  Medical units positioned at zone edges
-  Helicopters prioritized for blocked access areas

Results:

-  Focused effort on areas most likely to have survivors
-  Faster response times
-  More efficient use of limited equipment

“The damage map changed everything. We knew exactly where the worst destruction was”— Turkish Search and Rescue Team



Beyond Immediate Rescue - Long-Term Impact



Optimization & Simulation Lab

WEEK 1: SEARCH AND RESCUE

- ✓ Prioritize rescue operations
- ✓ Direct international aid teams
- ✓ Allocate heavy equipment

WEEK 1-2: EMERGENCY SHELTER

- ✓ Estimate number of displaced people
- ✓ Plan shelter locations
- ✓ Avoid unsafe areas

WEEK 2-4: INFRASTRUCTURE ASSESSMENT

- ✓ Identify damaged hospitals
- ✓ Map blocked roads for debris removal
- ✓ Assess critical facilities (power, water)

MONTH 2-3: DAMAGE DOCUMENTATION

- ✓ Insurance claims processing
- ✓ Government compensation programs
- ✓ Economic impact assessment

MONTH 6+: RECONSTRUCTION PLANNING

- ✓ Urban planning decisions (where to rebuild)
- ✓ Building code improvements
- ✓ Seismic risk mapping

Ongoing: "CNN damage maps serve as baseline for monitoring reconstruction progress"

A single damage assessment, completed in 72 hours, informed decision-making for months afterward. The CNN didn't just help with rescue—it created a foundation for the entire disaster response and recovery

Real-World Challenges - What Made This Difficult

DATA GAPS

- ❑ **Cloud cover** delayed imagery for some areas (SAR interpretation is more complex for AI models). Some regions waited 72+ hours for clear satellite images
- ❑ Many regions **lacked high-quality pre-event imagery** or accurate building footprint datasets
- ❑ AI models (e.g., xView2, CNN-based approaches) were **pre-trained** on datasets like xBD, which may **not fully represent Turkey's building styles** and earthquake damage patterns.
- ❑ Uploading and processing **large imagery datasets required stable internet** and compute resources, which were limited in disaster zones.
- ❑ Dense urban areas with **collapsed debris and shadows made segmentation harder for AI models**, leading to false negatives or misclassification
- ❑ **Aftershocks continued**, changing damage over time

Real-World Challenges - What Made This Difficult

MODEL CONFIDENCE & TRUST

- Emergency managers initially skeptical of AI, "How do we know the CNN is right?"
- Reluctance to deploy resources based on algorithms

INTERNET & INFRASTRUCTURE

- Internet disrupted in affected areas
- Power outages limited connectivity
- Cloud-based tools inaccessible to field teams

RESOURCE CONSTRAINTS

- CNN processing required expensive cloud computing (GPUs)
- Cost: \$2,000-5,000 per city for compute resources
- Smaller disasters or poorer countries may not afford this

Progress Since Turkey - The Technology Is Improving

1. Faster Processing

- 2024 Morocco Earthquake: Damage maps in 12 hours (6× faster!)
- Improved algorithms + pre-positioned models

2. Multiple Data Sources

- Optical satellites + Radar (SAR) = works through clouds
- Social media posts analyzed for damage reports
- Combines AI assessment with crowd-sourced information

3. Drone Deployment

- CNNs running on drones for immediate local assessment
- 5cm resolution (vs. 50cm from satellites)
- Can fly within hours of disaster, no waiting for satellites

2023 Turkey	2024+ Future
72-96 hour wait	12-24 hour wait
Optical and SAR	Multi-source fusion
Cloud computing	Edge AI (runs anywhere)
City-wide assessment	Building-by-building detail

This technology is evolving rapidly. What took 72 hours in 2023 might take 12 hours by 2025. The trajectory is clear: faster, more accurate, more accessible



MAKING AI DAMAGE ASSESSMENT AVAILABLE WORLDWIDE

PILLAR 1: SHARED DATASETS

- xBD dataset expanded with Turkey earthquake data
- Now includes 20+ disaster types from 6 continents
- Freely available for researchers and emergency agencies
- **Why It Matters:** "Better training data = models that work in more countries and building types"

PILLAR 2: GLOBAL REPOSITORIES

Organizations Building AI Tools:

- **United Nations OCHA:** Disaster response AI toolkit
- **World Bank:** Damage assessment protocols for developing countries
- **European Space Agency:** Free satellite imagery for disasters

PILLAR 3: TRAINED PEOPLE

The Human Element:

- Technology alone isn't enough
- Need emergency managers who understand AI
- Need AI researchers who understand disasters

Your Role: "The next generation of disaster response will be AI-literate. You are part of building that capability."

Key Takeaways

1. SPEED SAVES LIVES

CNNs reduced damage assessment time from 7-14 days to 3-4 days in Turkey. That kept assessments within the critical 72-hour rescue window.

2. BETTER DECISIONS WITH DATA

Damage maps enabled targeted resource deployment. ~75% of rescue teams focused on 47 priority zones, saving many hours of search time.

3. TECHNOLOGY IS NOT PERFECT

80% accuracy means 20% errors. Challenges included data gaps, model trust, infrastructure failures, and costs. Human oversight remains essential.

4. AI + HUMANS = BEST RESULTS

~88% of buildings auto-classified, ~12% reviewed by experts. AI handles the obvious cases, humans handle the uncertain ones. This is decision support, not replacement.

5. YOU ARE THE NEXT GENERATION

Disaster response needs AI-literate emergency managers and disaster-aware AI researchers. The skills you're learning this week are directly applicable to saving lives.

Discussion - Your Turn to Think

Design Challenge: Damage Assessment for YOUR City/Country: If you were designing a CNN-based damage assessment system for YOUR city/country, what would be your top 3 priorities?

- **LOCAL BUILDING TYPES** (*Does your country look like Turkey, or very different?*)
 - What construction is common? (wood, concrete, masonry, mixed)
 - Urban vs. rural buildings
 - Historical vs. modern construction
- **AVAILABLE DATA & RESOURCES** (What data can you realistically access in an emergency?)
 - Satellite imagery access (free? paid? resolution?)
 - Historical building data (maps, databases)
 - Computing infrastructure (cloud access? local GPUs?)
 - Internet reliability during disasters
- **DISASTER TYPES & PRIORITIES** (What disaster scenario are you preparing for?)
 - Most common hazards (earthquakes, floods, hurricanes, wildfires?)
 - Population density patterns
 - Response time requirements



RESOURCES FOR FURTHER LEARNING

- Gupta et al. (2019): "xBD: A Dataset for Assessing Building Damage"
- Gholami, S., Robinson, C., Ortiz, A., Yang, S., Margutti, J., Birge, C., ... & Ferres, J. L. (2022, November). On the deployment of post-disaster building damage assessment tools using satellite imagery: A deep learning approach. In 2022 IEEE International Conference on Data Mining Workshops (ICDMW) (pp. 1029-1036). IEEE.
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- Buznego Puerto, C., van den Homberg, M. J. C., Margutti, J., Kundak, S., Goksu, C., Yilmaz, A., ... & Kerle, N. (2023, November). Complementarity of Field and Remote Sensing Based Damage Assessments in the Early Analysis of Shelter Loss for the February 2023 Türkiye Earthquakes. In International Conference on Energy and Environmental Science (pp. 477-489). Cham: Springer Nature Switzerland.
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- Hakami, A., Kumar, A., Shim, S. J., & Nahleh, Y. A. (2013). Application of soft systems methodology in solving disaster emergency logistics problems. International Journal of Mechanical, Aerospace, Industrial, Mechatronic and Manufacturing Engineering, 7(12), 2470-2477.
- UN OCHA: Disaster response resources
- EERI (Earthquake Engineering Research Institute): Field reconnaissance reports
- EMSR648 - Rapid Mapping: Earthquake in East Anatolian Fault Zone



Thank you!



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